When Do Answers Change? Estimating Question Recency Demands in QA

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Abstract

Large language models (LLMs) answer questions using knowledge that may change over time; however, most evaluations remain static. This work explores the demands of question recency, specifically the frequency with which answers must be updated to maintain accuracy, and the precision with which models can predict the necessary update frequency. We construct a *generation-and-labeling* pipeline that produces recency-aware questions along two axes: stationary vs.versus non-stationary temporal behaviour, and single-event vs. multi-event dependencies. The multi-event category captures questions whose answers rely on multiple temporally connected events, including causal and purely co-occurring pairs. The single-event category focuses on questions about a single evolving process. Stationarity labels distinguish predictable update cycles from volatile changes. Employing a Llama model, the pipeline generates and labels a 600-question dataset, assigning recency classes, stationarity labels, and multi-event markers. Subsequently, it applies perturbations to construct more challenging temporal variants. The resulting dataset extends prior work, providing new and more demanding questions with which to evaluate LLMs on when those answers change.

1 Introduction

In the last couple of years, large language models (LLMs) have demonstrated remarkable capabilities across a wide range of natural language processing tasks. In particular, their performance on question answering (QA) tasks has been studied extensively [Fischer et al., 2024, Brown et al., 2020], with very strong results. However, a persistent challenge for QA systems is handling temporal dynamics, specifically answering questions whose correct answers change over time.

Much of this prior work largely ignores the fact that many real-world questions require information that is time-sensitive and may change frequently, such as:

- *"What is the current population of Germany?"*
- *"Who is the current president of the United States?"*

As facts evolve, answers that were once correct inevitably degrade, and outdated responses can harm user trust. Despite this, many QA evaluations remain static, implicitly assuming that factual correctness is invariant over time.

Prior work on temporal QA and time-aware modeling has highlighted this limitation from several perspectives. These include datasets for time-sensitive questions [Chen et al., 2021], temporal knowledge bases for language models [Dhingra et al., 2022], real-time QA benchmarks [Kasai et al., 2024], search-augmented approaches for refreshing model knowledge [Vu et al., 2024], and self-updating present-anchored QA suites [Meem et al., 2024]. While these efforts underscore the importance of temporal awareness, they typically treat time sensitivity as a coarse or binary property and do not explicitly characterize how frequently answers change or how complex their temporal dependencies are.

Dataset	Creation	KC	Multi-hop	Recency-Label	Multi-Event	# Ques.
TimeQA [Chen et al., 2021]	Templ.-Wikidata	Wikipedia	✗	✗	✗	20 000
SituatedQA [Zhang and Choi, 2021]	Man.-Filt.	Wikipedia	✗	✗	✗	12 000
TempLama [Dhingra et al., 2022]	Templ./Cloze	Custom-News	✗	✗	✗	50 000
StreamingQA [Liška et al., 2022]	Man.+Gen.	WMT News	✓	✗	✗	410 000
ArchivalQA [Wang et al., 2022]	Gen.	NYT Articles	✗	✗	✗	532 000
ChroniclingAmericaQA [Piryani et al., 2024]	Gen.	Chronicling America Newspapers	✗	✗	✗	485 000
RealTimeQA [Kasai et al., 2024]	News websites	News Articles	✓	✗	✗	∼5000
PATQA [Meem et al., 2024]	Templ.-wikidata	Wikipedia	✓	✗	✗	6172
FreshQA [Vu et al., 2024]	Man.	Google Search	✓	✗	✗	600
RecencyQA (*unpublished*)	Man.-Filt.+Gen	Wikipedia/Wikidata	✓	✓	✗	6115
RecencyQA-Multi (*ours*)	Man.-Filt.+Gen	RecencyQA+LLM	✓	✓	✓	1432

Table 1: Overview of question answering datasets. Abbreviations: *Man.*=created manually, *Gen.*=Automatically generated, *Man.-Filt.*=filtered from other datasets, *Man.+Gen.*=created by crowdsourcing and LLM generation *Templ.*=created using templates, *Man.-Filt.+Gen*=filtered from other datasets and LLM generation, *KC*=Knowledge Corpus.

We refer to this requirement as the *recency demand* of a question: the degree to which an answer must be updated over time to remain accurate. Recency demand varies across questions, depending on factors such as the predictability of change and whether an answer depends on a single evolving event or on multiple temporally related events.

A very recent paper, *RecencyQA* (unpublished), introduces a dataset explicitly labeled with recency demands. The authors annotate questions according to how frequently their answers are expected to change and use this dataset to evaluate the ability of various LLMs to predict a question's recency demand. Their results indicate that, while models capture coarse distinctions between static and rapidly changing facts, accurately estimating finer-grained recency requirements remains challenging.

Building on this line of work, our approach extends prior recency-aware QA datasets along several important dimensions. Rather than relying on a fixed set of annotated questions, we introduce a generation-and-labeling pipeline that systematically expands a small seed set into a substantially richer collection of temporal questions. Starting from 75 input questions, the pipeline produces approximately 1500 recency-aware questions by varying both temporal behavior and event structure. In addition to distinguishing stationary from non-stationary temporal dynamics, we explicitly control for event dependency—an aspect that has received little explicit attention in prior work—by generating single-event questions as well as multi-event questions involving two or three temporally connected events, with both causal and non-causal relationships. Each generated question is further annotated with one or more recency labels and contextual conditions under which the label applies. This structured expansion enables finer-grained analysis of temporal complexity in QA than prior datasets, which typically focus on isolated questions or coarse recency distinctions.

We start explaining the generation pipeline in Section 2, followed by testing various LLMs on the created dataset in Section 3. Using the results from the experiment, we fine-tune a smaller model and compare it to the larger LLMs in Section 4. Finally, we conclude with a discussion of findings and future work in Section 5.

2 Dataset Creation

Our dataset creation is closely aligned with the *RecencyQA* paper's methodology, but we extend their approach by incorporating multi-event questions and refining the stationarity aspect. The dataset generation pipeline consists of two main stages: question generation and recency labeling, as illustrated in Figure 1. This pipeline constructs (as described in the following sections) a diverse set of recency-aware questions, each annotated with appropriate recency labels

Recency Class	Expected time until answer change	Example Question
An-Hour	Within an hour	What is the current stock price of Apple?
A-Few-Hours	Within a few hours	What is the current traffic situation on the A9 highway?
A-Day	Within a day	What is today's weather forecast for Munich?
A-Few-Days	Within a few days	What movies are currently trending on Netflix this week?
A-Week	Within a week	What are the top-ranked songs on the Billboard chart this week?
A-Few-Weeks	Within a few weeks	What is the current FIFA world ranking of the German national team?
A-Month	Within a month	What is the current unemployment rate in Italy?
A-Few-Months	Within a few months	What is the current inflation rate in the Eurozone?
A-Year	Within a year	What is the current version of the Java programming language?
A-Few-Years	Within a few years	Who is the president of the United States?
Many-Years	After many years	What is the population of Germany?
Never	Never changes	What is the chemical symbol for gold?

Table 2: The recency classes proposed by the *RecencyQA* paper, along with example questions for each class.

and contextual conditions. The code and dataset are publicly available at GitHub[1] for reproducibility and further research.

2.1 Question Taxonomy

As a core component of the dataset generation pipeline, we organize questions along a small set of structural dimensions, which are stored as explicit metadata for every record.

- **Stationarity**: Questions differ in whether their correct answers are stable over time or change as the underlying real-world situation evolves. Non-stationary questions require access to recent information, while stationary questions remain valid independently of time.

- **Multi-Event**: Questions may depend on a single situation or combine information from multiple distinct or connected events. Multi-event questions increase temporal and structural complexity by requiring the integration of information across several concurrent or related processes.

- **Inter-event relationship**: When multiple events are involved, their relationship can be causal—where one event influences another—or purely temporal, where events co-occur in time without an assumed dependency.

- **Recency classes**: Each question is associated with a recency class indicating the expected timescale on which its answer may change (e.g., hours, days, or years). We adopt the recency classes proposed by the *RecencyQA* paper (see Table 2).

- **Contextual conditions**: Recency is not absolute but depends on the assumed state of the world. Contextual conditions describe the situation under which a given recency class applies, allowing the same question (under Non-Stationary) to have different recency labels.

2.2 Question Selection

Having defined the question taxonomy, the next step in the pipeline is to select an initial set of seed questions from which new items are generated. These seeds serve as prompts that anchor topic, style, and temporal structure while still allowing the language model to introduce new events, domains, and cross-event interactions.

In our setting, we randomly select 75 questions from the original *RecencyQA* dataset as seed inputs. The selected questions are evenly distributed across stationarity and recency classes. All questions are provided to the model in a structured JSON format (see Appendix A.1), which allows the generator to preserve key properties of the original question while flexibly expanding it into new single- and multi-event formulations.

[1] GitHub | RecencyQA_MultiEvent

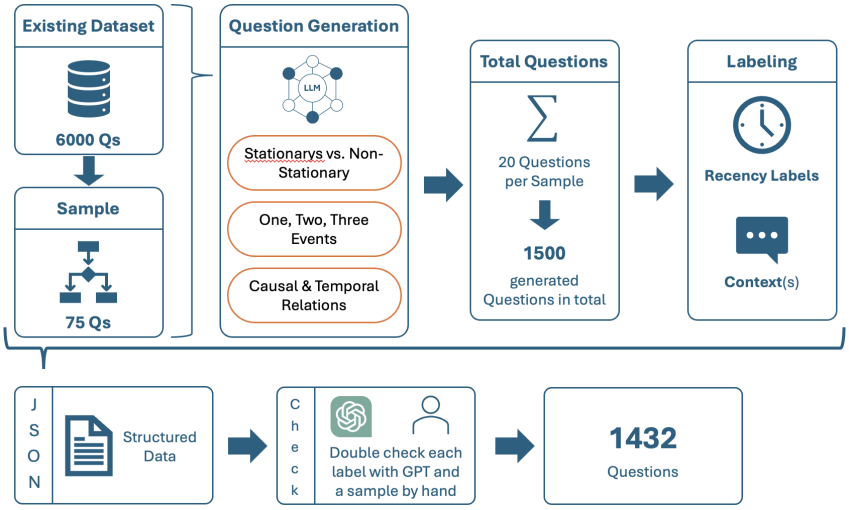

Figure 1: Caption text.

2.3 Question generation

Prompt templates instruct the model to output exactly two questions per call and forbid placeholders or paraphrasing of the seed (Appendix B.1). A helper repeats calls until the target count is reached for each mode, reissuing only the missing portion to avoid over-generation. The ten prompt families are: stationary single (Appendix B.1.1); stationary two-event causal (Appendix B.1.2) and temporal-only (Appendix B.1.3); stationary three-event causal (Appendix B.1.4) and temporal-only (Appendix B.1.5); non-stationary single (Appendix B.1.6); non-stationary two-event causal (Appendix B.1.7) and temporal-only (Appendix B.1.8); and non-stationary three-event causal (Appendix B.1.9) and temporal-only (Appendix B.1.10). Every generated question is tagged with the seed identifier and metadata for stationarity, event dependency, number of events, and (when applicable) whether the multi-event relation was prompted as causal or temporal-only.

2.4 Recency labeling

Each question is labeled with recency classes that describe how quickly its answer should be refreshed. We use the following twelve labels:

An hour	A few hours
A day	A few days
A week	A few weeks
A month	A few months
A year	A few years
Many years	Never

Stationary questions receive one label; non-stationary questions receive two labels to capture alternative temporal regimes. For every label we also elicit a one-sentence temporal context (event, phase, or condition) that makes the question relevant. The labeling prompt prohibits meta-reasoning or explicit time stamps and is reproduced in Appendix B.2 via the stationary template (Appendix B.2.1) and the non-stationary template (Appendix B.2.2). Labels and contexts are merged into a structured list so downstream models can pair each recency class with its motivating scenario. We retain only entries whose labels parse as JSON and whose context count matches the expected label count.

2.5 Sampling and size

Running the ten-generation-mode configuration on 75 seed questions yields 1,498 retained questions after discarding malformed generations that fail JSON parsing or lack labels. The ensure-and-top-off loop keeps modes balanced: 748 stationary vs. 750 non-stationary questions; 300 single-event, 598 two-event, and 600 three-event items; and

within multi-event items 600 were prompted as causal chains and 598 as temporal-only pairs or triples. Mode totals differ from the ideal 1,500 by two items that were filtered for malformed JSON.

2.6 Output format

All examples are written as one JSON object per line. Each record stores the seed question identifier, question text, stationarity flag, event dependency type, number of events, the causal vs. temporal-only generation tag for multi-event items, and the list of recency labels with their paired contexts. The resulting corpus offers controlled variation across temporal behaviors and provides explicit supervision for estimating when an answer is likely to drift. We also keep the original seed linkage so downstream work can group generated questions by provenance or remove seed-specific leakage.

2.7 Dataset Statistics

The released split contains 1,498 examples. Stationarity is balanced (748 stationary, 750 non-stationary), and label cardinality follows the prompt contract (one label for stationary, two for non-stationary). Event cardinality is spread across 300 single-event, 598 two-event, and 600 three-event questions, with multi-event items split nearly evenly between causal (600) and temporal-only (598) prompting. The most common recency labels are "A-Year" (406), "A-Day" (368), and "A-Few-Days" (341); the tail includes a handful of model-emitted variants (e.g., "A-Quarter", "A-Few-Minutes") that remain in the data for transparency and can be remapped if a downstream system requires a closed label set.

3 Experiments

4 LLM Fine-Tuning

5 Conclusion and Future Work

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A JSON-structures

A.1 Input JSON format for the pipeline

```
[
  {
    "q_id": "<string | int>",
    "question": "<string>"
  }
]
```

Figure 2: JSON structure for the RecencyQA_MultiEvent pipeline input.

A.2 Output JSON format from the pipeline

```
[
  {
    "q_id": "<string | int>",
    "question": "<string>",

    "stationarity": "<'Stationary' | 'Non-Stationary'>",
  }
]
```

Figure 3: JSON structure for the RecencyQA_MultiEvent pipeline output.

B Prompt Templates

B.1 Generation prompts

B.1.1 Stationary single-event

```
You generate STATIONARY temporal questions.

Definition (internal):
- A stationary question has a stable temporal update behavior.
- The answer changes over time, but the timespan how often the answer must be updated
  would remain the same regardless of when the question is asked.

Task:
Generate EXACTLY 2 stationary temporal questions focusing on ONE event or process.

Rules:
- Must rely on real-world phenomena that change over time.
- Must be stable, cyclical, predictable, or rhythm-based.
- Must NOT paraphrase the examples.
- Must NOT use placeholders.
- Do NOT mention stationarity in the question.

Examples:
{examples}

Return JSON:
{
  "questions": ["q1","q2"]
}
```

B.1.2 Stationary multi-event (causal)

```
You generate STATIONARY multi-event temporal questions.

Task:
Generate EXACTLY 2 stationary temporal questions involving TWO events that are causally
↪  related.

Rules:
- Must involve at least TWO distinct temporal events.
- Events must have a clear cause-effect relationship.
- Temporal behavior must be stable, predictable, cyclical, or regular.
- Must rely on real-world change.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention stationarity.

Examples:
{examples}

Return JSON:
{
  "questions": ["q1","q2"]
}
```

B.1.3 Stationary multi-event (temporal-only)

```
You generate STATIONARY multi-event temporal questions.

Task:
Generate EXACTLY 2 stationary temporal questions involving TWO events that are ONLY
↪  temporally connected.

Rules:
- Events must be from clearly different real-world domains.
- Events must NOT influence each other causally.
- Must NOT belong to the same topic, organization, or event series.
- Temporal behavior must be stable, predictable, cyclical, or regular.
- Must rely on real-world change.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention stationarity.

Examples:
{examples}

Return JSON:
{
  "questions": ["q1","q2"]
}
```

B.1.4 Stationary three-event (causal)

```
You generate STATIONARY multi-event temporal questions.

Task:
Generate EXACTLY 2 stationary temporal questions involving THREE events
that are causally related in a chain or network.

Rules:
- Must involve EXACTLY THREE distinct temporal events.
- Events must have clear cause-effect relationships.
- Temporal behavior must be stable, predictable, cyclical, or regular.
- Must rely on real-world change.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention stationarity.

Examples:
```

```
{examples}

Return JSON:
{
  "questions": ["q1", "q2"]
}
```

B.1.5 Stationary three-event (temporal-only)

```
You generate STATIONARY multi-event temporal questions.

Task:
Generate EXACTLY 2 stationary temporal questions involving THREE events
that are ONLY temporally connected.

Rules:
- Must involve EXACTLY THREE distinct events.
- Events must be from clearly different real-world domains.
- Events must NOT influence each other causally.
- Must NOT belong to the same topic, organization, or event series.
- Temporal behavior must be stable, predictable, cyclical, or regular.
- Must rely on real-world change.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention stationarity.

Examples:
{examples}

Return JSON:
{
  "questions": ["q1", "q2"]
}
```

B.1.6 Non-stationary single-event

```
You generate NON-STATIONARY temporal questions.

Definition (internal):
- A non-stationary question has unstable temporal update behavior.
- How frequently the answer must be updated depends strongly on WHEN the question is asked.
- OR the question is only relevant within short, event-dependent windows.

Task:
Generate EXACTLY 2 non-stationary temporal questions focusing on ONE event.

Rules:
- Must rely on quickly evolving or unstable processes.
- Must be meaningful and real-world.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention non-stationarity explicitly.

Examples:
{examples}

Return JSON:
{
  "questions": ["q1","q2"]
}
```

B.1.7 Non-stationary multi-event (causal)

```
You generate NON-STATIONARY multi-event temporal questions.

Task:
Generate EXACTLY 2 non-stationary temporal questions involving TWO events that are causally
↪  related.
```

```
Rules:
- At least one event must be unstable, unpredictable, or highly time-sensitive.
- Events must have a clear cause-effect relationship.
- Must rely on real-world temporal change.
- Must involve at least TWO distinct temporal events.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention non-stationarity.

Examples:
{examples}

Return JSON:
{
  "questions": ["q1","q2"]
}
```

B.1.8 Non-stationary multi-event (temporal-only)

```
You generate NON-STATIONARY multi-event temporal questions.

Task:
Generate EXACTLY 2 non-stationary temporal questions involving TWO events that are ONLY
↪  temporally connected.

Rules:
- Events must be from clearly different real-world domains.
- Events must NOT influence each other causally.
- Must NOT belong to the same topic, organization, or event series.
- At least one event must be unstable, unpredictable, or highly time-sensitive.
- Must rely on real-world temporal change.
- Must involve at least TWO distinct temporal events.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention non-stationarity.

Examples:
{examples}

Return JSON:
{
  "questions": ["q1","q2"]
}
```

B.1.9 Non-stationary three-event (causal)

```
You generate NON-STATIONARY multi-event temporal questions.

Task:
Generate EXACTLY 2 non-stationary temporal questions involving THREE events
that are causally related.

Rules:
- Must involve EXACTLY THREE distinct temporal events.
- At least one event must be unstable, unpredictable, or highly time-sensitive.
- Events must have clear cause-effect relationships.
- Must rely on real-world temporal change.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention non-stationarity.

Examples:
{examples}

Return JSON:
{
```

```
  "questions": ["q1", "q2"]
}
```

B.1.10 Non-stationary three-event (temporal-only)

```
You generate NON-STATIONARY multi-event temporal questions.

Task:
Generate EXACTLY 2 non-stationary temporal questions involving THREE events
that are ONLY temporally connected.

Rules:
- Must involve EXACTLY THREE distinct temporal events.
- Events must be from clearly different real-world domains.
- Must NOT influence each other causally.
- At least one event must be unstable, unpredictable, or highly time-sensitive.
- Must rely on real-world temporal change.
- Do NOT paraphrase examples.
- Do NOT use placeholders.
- Do NOT mention non-stationarity.

Examples:
{examples}

Return JSON:
{
  "questions": ["q1", "q2"]
}
```

B.2 Labeling prompts

B.2.1 Stationary labeling

```
Analyze this temporal question and produce temporal labels:

"{question}"

Your tasks (internal reasoning only, output JSON only):

1. Provide ONE recency label:
{
  "An-Hour": "changes within one hour",
  "A-Few-Hours": "changes within a few hours",
  "A-Day": "changes within one day",
  "A-Few-Days": "changes within a few days",
  "A-Week": "changes within one week",
  "A-Few-Weeks": "changes within a few weeks",
  "A-Month": "changes within one month",
  "A-Few-Months": "changes within a few months",
  "A-Year": "changes within one year",
  "A-Few-Years": "changes within a few years",
  "Many-Years": "changes after 10 or more years",
  "Never": "never changes"
}

2. Based on the selected label, write a short temporal context (ONE sentence) describing the
↪  current event, phase, or condition in which the question is asked. This does not have to
↪  be related to the question causally.
   - Must describe an EVENT, PHASE or CONDITION, taking place when the question becomes
   ↪  relevant.
   - No specific years.
   - No meta reasoning.
   - No "current"
   - No "the question is asked"

Return JSON:
{
  "recency_list": ["label1"],
```

```
  "context_list": ["context1"]
}
```

B.2.2 Non-stationary labeling

```
Analyze this temporal question and produce temporal labels:

"{question}"

Your tasks (internal reasoning only, output JSON only):

1. Provide TWO recency labels (for two different realistic temporal situations), choose from:
{
  "An-Hour": "changes within one hour",
  "A-Few-Hours": "changes within a few hours",
  "A-Day": "changes within one day",
  "A-Few-Days": "changes within a few days",
  "A-Week": "changes within one week",
  "A-Few-Weeks": "changes within a few weeks",
  "A-Month": "changes within one month",
  "A-Few-Months": "changes within a few months",
  "A-Year": "changes within one year",
  "A-Few-Years": "changes within a few years",
  "Many-Years": "changes after 10 or more years",
  "Never": "never changes"
}

2. For each selected label provide ONE short temporal context (ONE sentence each) describing
↪  the current event, phase, or condition in which the question is asked. This does not have
↪  to be related to the question causally.
  - Each context must describe a different EVENT, PHASE or CONDITION, taking place when the
  ↪  question becomes relevant.
   - No specific years.
   - No meta reasoning.
   - No "current"
   - No "the question is asked"

Return JSON:
{
  "recency_list": ["label1", "label2"],
  "context_list": ["context1", "context2"]
}
```